Pratham Satani

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Education

M.S. in Artificial Intelligence — **Sep 2025 – Dec 2027**
Northeastern University — Boston, MA, USA

B. Tech. in Information Technology — **Oct 2021 – Jun 2025**
Gujarat Technological University — India

Skills

Programming & Databases: Python, C++, TypeScript, Chroma, Milvus, FAISS, Pinecone, NoSQL, SQL
Machine Learning & Data Science: TensorFlow, PyTorch, Keras, Scikit Learn, Statsmodels, HuggingFace, NumPy, Pandas
MLOps & Deployment: Github Actions, MLflow, Docker, Kubernetes, DVC, CI/CD, Flask, FastAPI, Postman
Generative & Agentic AI: LangChain, LangGraph, LangSmith, RAG, Prompt Engineering, LLMs
Cloud Platforms: GCP (Vertex AI, Cloud Run, BigQuery, Compute Engine), AWS (BedRock, SageMaker, DynamoDB, EC2)

Experience

Data Scientist Co-op — **Jun 2026 – Present**
Northeastern University — Portland, ME, USA

- Building an **AI-native Digital Risk Protection** pipeline that detects malicious content targeting the client's customers, aimed at improving and potentially replacing their current system
- Engineering multimodal **data and feature pipelines** across text, image, and video, generating embeddings and matching them against relational stores and vector indexes **at scale**
- **Training and fine-tuning** models to a target precision/recall bar, then **deploying** them to production with an automated evaluation and drift-monitoring loop for stable quality
- Built an agentic AI **data generation** system producing a balanced multimodal benchmark of 50 enterprise profiles and ∼1,400 items per incident type for **model training and evaluation**

AI Engineer Intern — **Jan 2025 – Jun 2025**
Softmax AI LLC — India

- Collaborated with the AI engineering team to design a scalable **agentic AI automation system** using **FastAPI**, **LangChain**, and **ChromaDB**, enabling autonomous handling of customer support workflows and reducing ticket handling time
- Designed and implemented a production-grade **RAG-based AI agent** that combined dense retrieval with LLM reasoning to autonomously query and synthesize information from a **5,200 document** knowledge base
- Built and deployed a modular, containerized **LLM agent backend** on **GCP (Cloud Run)** with **CI/CD pipelines**, achieving **95% retrieval accuracy** and sub-1s response latency for real-time responses
- Reduced repetitive customer support workload by **80%** by developing a reusable **agentic RAG framework** that enabled intelligent task execution and was adopted across multiple client automation systems

Machine Learning Engineer — **Jan 2024 – Dec 2024**
CasePoint LLC — India

- Collaborated with cross-functional teams to design and deploy ML-powered cloud solutions for enterprise automation, with a focus on **Computer Vision** and **Recommendation Systems**
- Designed and deployed a multimodal apparel recommendation system leveraging **fashionSigLIP** and **Vertex AI Vector Search**, combining product images with auto-generated textual descriptions for improved retrieval
- Built scalable data ingestion and multimodal embedding pipelines processing **1.6M products**, with modular FastAPI services, containerized deployments on GCP, and CI/CD-enabled production workflows

Academic Projects

Links to all Projects

Web-Scale Retrieval-Augmented Generation System — Link to GitHub

- Architected an end-to-end RAG system, including web-scale data ingestion, text chunking, embedding generation, and vector similarity search, enabling grounded LLM responses over large unstructured knowledge sources.
- Developed scalable ML-backed APIs using FastAPI, integrating LLM inference, retrieval pipelines, and PostgreSQL metadata storage, with containerized deployment via Docker for reproducibility and production readiness.
- Optimized retrieval accuracy and latency by tuning **embedding models**, chunking strategies, and **k-NN search parameters** in ChromaDB, applying ML evaluation and profiling techniques to improve answer relevance and system throughput.

KrishiMitra: ML-driven Crop Recommendation System — Link to GitHub

- Trained and evaluated supervised machine learning models including **Random Forest** and **XGBoost**, achieving **92% accuracy** using cross-validation and performance metrics